KLH

UNIVERSITY · HYDERABAD

PROJECT PRESENTATION

Department of Computer Science and Engineering · KLH Bachupally Campus, Hyderabad

TEAM NUMBER

GUIDE NAME

Dr. Jyothsna Datti

PROJECT TITLE

Heart Disease Prediction Using Logistic Regression, SVM, and KNN

INTRODUCTION

- **Overview:** Prognosis tool evaluating cardiovascular risk from 13 clinical indicators (1,025 records).
- **Problem:** Delayed diagnosis and invasive clinical procedures increase patient mortality.
- **Objectives:** Benchmark LR, SVM, and KNN to identify high-risk individuals accurately.
- **Significance:** Enables rapid, accessible point-of-care clinical decision support with interpretable outputs.

RESEARCH GAP

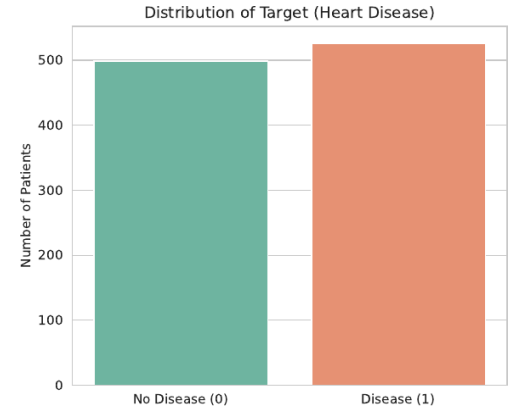
- **Limitations:** Traditional heuristics rely on rigid single-variable thresholds with high false negatives.
- **Unexplored:** Cross-comparison of margin vs instance classifiers on balanced clinical data.
- **Need:** Interpretable probability scoring combined with strict non-linear margin classification.
- **Our Solution:** End-to-end pipeline: robust scaling, 3-model benchmark & real-time patient web portal.

ALGORITHM

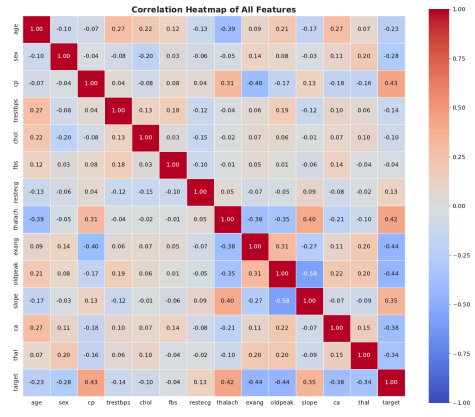


RESULTS

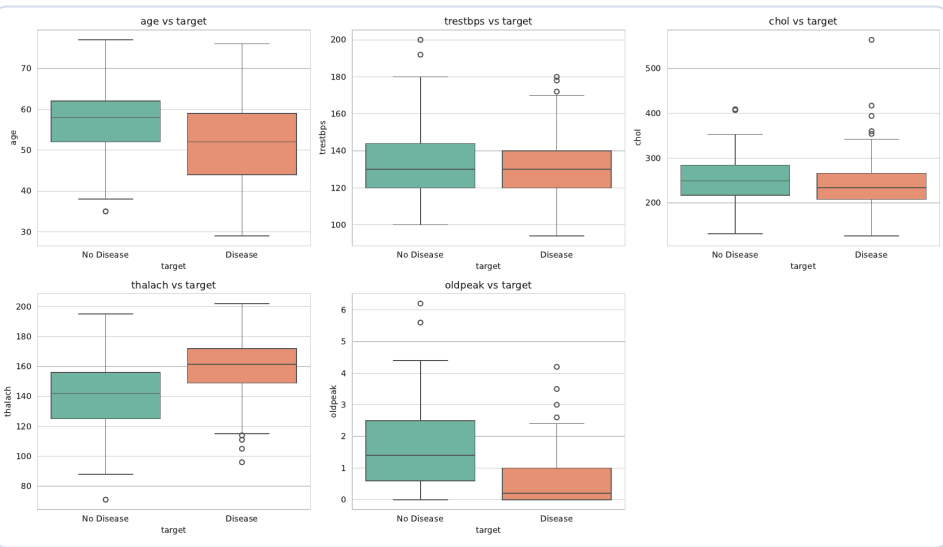
Target distribution



Correlation heatmap of all features



Numerical features vs target



Categorical features vs target

